



Effect of phase-scale regulation on electromagnetic shielding effectiveness of co-continuous PP/PBT/CNTs composites

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ABSTRACT

Herein, the carbon nanotubes (CNTs)-induced polymer-phase scale behavior was regulated in polypropylene/poly(butylene terephthalate) (PP/PBT) composites. To determine the effects of CNTs migration from the PP to the PBT phase on phase viscosity, morphology, and domain scale, PP/PBT/CNTs composites were prepared and systematically analyzed using experimental characterizations and numerical simulations. During composite preparation, the viscosity of both phases transited, the domain scale was refined from 3.1 μm to 1.8 μm , and interphase compatibility was significantly improved. When the CNTs content reached 3 vol%, the migration behavior became saturated. Furthermore, a co-continuous structure with double percolation, as along with more conductive pathways and complex interfaces, was formed in the PP/PBT/CNTs composites, leading to performance stabilization. Consequently, the prepared PP/PBT/CNTs composites exhibited a conductive percolation threshold of 0.19 vol% CNTs and an electromagnetic interference (EMI)-shielding effectiveness (SE) of ~ 36 dB, accompanied by significant reflection and absorption losses. Furthermore, to further regulate phase morphology, compatibilizers such as maleic anhydride grafted PP (PP-g-MAH), glycidyl methacrylate grafted PP (PP-g-GMA), and glycidyl methacrylate grafted polyolefin elastomer (POE-g-GMA) were incorporated, further confirming the correlation between phase-domain size and EMI-SE. Overall, this study demonstrates a strategy for designing eco-friendly nanocomposites with both ultralow percolation thresholds and excellent EMI-SE.

1. Introduction

With the rapid development of information technology, electronic devices have become increasingly widespread, and severe electromagnetic interference (EMI) and radiation have raised growing concern because of their inevitable harm to the environment and human health, as well as their adverse effects on information security and equipment reliability [1–4]. Compared with metal-based EMI-shielding materials, conductive polymer composites (CPCs) are lighter, less expensive, and exhibit superior corrosion resistance and processability [5,6]. Traditionally, CPCs have been fabricated by incorporating conductive functional fillers (e.g. carbon black (CB) [7–9], carbon nanotubes (CNTs) [5,10], graphene oxide (GO) [11], metallic powders [12] or liquid metals [4,13,14], etc.) into highly insulating polymer matrices. However, at low conductive filler contents, most conventional CPCs fail to meet commercial EMI-shielding requirements (>20 dB). Therefore, numerous structures have been developed to achieve high EMI-SE for CPCs [15,16], such as segregated [17,18], layered [19,20], and foaming

structures [21], among which the construction of co-continuously structured polymer composites has attracted considerable research attention [22]. The incorporation of conductive fillers into co-continuously structured immiscible polymer blends can effectively reduce the conductive percolation threshold of CPCs through the double-percolation mechanism [23,24]. Theoretically, polymer composites with double-percolated conductive networks can exhibit high EMI-SE even at low filler contents, primarily through ohmic losses when electromagnetic waves interact with the conductive pathways. Moreover, at the phase interfaces of immiscible polymer blends, multiple reflections, scattering losses, and impedance mismatches of electromagnetic waves further contribute to enhancing EMI-SE [25–28]. Kundalwal et al. [29,30] used theoretical calculations and numerical simulations to quantify the EMI-SE of a multiwalled carbon nanotube (MWCNT)/polypyrrole (PPy) nanocomposite and found that, within the 4.2–8.2 GHz (C-band) range, the CNT–matrix interphase thickness and CNT content significantly affected SE by modulating CNT tunneling conductivity. The as-prepared 3-mm-thick PPy nanocomposite exhibited

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an SE of 75 dB, demonstrating excellent C-band EMI-SE suitable for satellite communication, Wi-Fi devices, weather radar systems, and related applications.

In immiscible polymer composites, the formation of a co-continuous structure is governed by the polymer viscosity [31,32], volumetric ratio [32], and related parameters. Notably, the directed migration and preferential localization of conductive fillers between polymer phases are determined by interfacial tension [33], phase viscosity [34], thermodynamic and kinetic factors [14,35] etc., all of which, in turn, influence the viscosity of each polymer phase. Furthermore, variations in polymer-phase viscosity can either coarsen or refine the phase morphology, thereby reconstructing conductive networks and interfaces within the polymer composites and significantly affecting their electromagnetic properties. By incorporating silicon dioxide (SiO₂) into the polylactide (PLA) phase, Liu et al. [23] constructed a double-percolation structure in a PLA/polycaprolactone (PCL) composite containing only 30 wt% PCL. The resulting composite exhibited an electrical conductivity of $4.57 \times 10^{-4} \text{ S m}^{-1}$ and an EMI-shielding effectiveness (SE) of $\sim 26 \text{ dB}$, despite containing only 3.0 wt% CNTs. Therefore, regulating the phase scale or viscosity through the directional migration of conductive fillers represents a promising strategy for developing CPCs with highly efficient EMI-SE.

In nanocomposites, although the migration of functional fillers has been systematically investigated in different polymer systems [13,34,36,37], to the best of our knowledge, few studies have combined numerical simulations and experimental characterization to analyze how the directional migration of conductive fillers regulates phase domains, particularly under different blending methods, filler contents, and CPC EMI-SE goals. Therefore, in this work, PP/PBT/CNTs composites were prepared using different blending approaches and CNTs contents, and the mechanisms governing polymer-phase morphology and CNTs-migration-induced co-continuous structure formation were experimentally characterized and systematically analyzed through numerical simulations. By exploiting CNTs migration, the polymer-phase viscosity was modified, the domain scale was significantly refined, and denser conductive networks were established within the composites. Moreover, we developed ultralow-percolation-threshold, double-percolation conductive polymer nanocomposites exhibiting excellent EMI shielding with high reflection and absorption efficiencies. Finally, to further clarify the relationship between phase-domain regulation and EMI-SE, compatibilizers were incorporated into the PP/PBT/CNTs composites, simultaneously enhancing both the EMI shielding performance and mechanical properties.

2. Experimental

2.1. Materials

Carbon nanotubes (CNTs, NC7000) were purchased from Nanocyl SA, Belgium, with an average length of 1.5 μm , an average diameter of 9.5 nm, and a density of 1.75 g/cm³. Polypropylene (PP, RP-340H) was purchased from LyondellBasell, Netherlands, with a density of 0.9 g/cm³ and a melt flow rate of 1.8 g/10 min (230 °C/2.16 kg). Poly(butylene terephthalate) (PBT, 6131 NC010) was purchased from DuPont, USA, with a density of 1.3 g/cm³ and a melt flow rate of 48 g/10 min (250 °C/2.16 kg). Xylene was purchased from Shanghai Titan Science and Technology Co, Ltd for etching PP phases. Maleic anhydride grafted PP (PP-g-MAH, P353) was purchased from Dow Chemical, USA, with a density of 0.91 g/cm³. Glycidyl methacrylate grafted PP (PP-g-GMA, SPG-02) and methacrylic acid glycidyl ester grafted polyolefin elastomer (POE-g-GMA) were purchased from Shanghai Jiayi Rong Polymer Co., Ltd.

2.2. Sample preparation

For reducing volatility and moisture content, PP and PBT resins were

dried in a vacuum oven at 80 °C and 120 °C, respectively, for 12 h. After drying, PP, PBT and CNTs were melt blended by a stacked-rotors (as shown in Fig.S1) mixer (homemade) with rotor speed of 80 rpm. The melting temperature was set at 250 °C. For preparing different samples, the volume ratio of PP and PBT was set as 8:2, 5:5 and 2:8, and the contents of CNTs were set as 0, 0.5 vol%, 1 vol%, 2 vol%, 3 vol%, 4 vol% and 5 vol%. Four different kinds of blending sequences were also used for preparing the composites: I. One-step method: PP, PBT and CNTs were pre-mixed and added into the double-rotors mixer at the same time. The mixing time was set as 720 s. The prepared composites were named as PP/PBT/CNTs; II. PP/CNTs masterbatch method: firstly, PP and CNTs were melt blended in the double-rotors mixer for 720 s to obtain PP-CNTs masterbatch. Then, PBT was added into PP-CNTs masterbatch for further mixing with 180 s. And the prepared composites were named as PP-CNTs/PBT; III. PBT/CNTs masterbatch method: firstly, PBT and CNTs were melt blended in the double-rotors mixer for 720 s to obtain PBT-CNTs. Then, PP were added into PBT-CNTs masterbatch for further mixing with 180 s. And the prepared composites were named as PBT-CNTs/PP; IV. Two-step method: firstly, PP and PBT were melt blended in the double-rotors mixer for 720 s to obtain PP-PBT masterbatch. Then, CNTs were added into PP-PBT masterbatch for further mixing with 180 s. And the prepared composites were named as PP-PBT/CNTs. The PP-CNTs/PBT was also selected to blend with different compatibilizers (PP-g-MAH, PP-g-GMA and POE-g-GMA) for 12 min to obtain the composites. All prepared samples were hot-compressed by a flat vulcanizing machine at 250 °C and 15 Mpa to obtain required shape and dimension.

2.3. Numerical simulation method

The simulation software Fluent was used for simulating mixing process of PP/PBT/CNTs composites.

2.3.1. Physical geometric model

To reduce the time of simulation calculations, the cross-sections of the mixing chamber and kneading block units were modeled in two dimensions using software SolidWorks, and the feeding port of the mixer was ignored, as shown in Fig.S1 (B). Gambit software with adaptive meshes was used to mesh the simplified two-dimensional geometric model. The details are shown in Supporting Information.

2.3.2. Mathematical model

In order to solve and calculate the phase scale evolution of the polymers, a multiphase flow model coupling Fluid Volume Function (VOF) method and Level Set (Level-set) method was constructed in this work to capture the phase interface using a smooth distance function. The corresponding solving equations (continuity equation, momentum equation, VOF control equation, and Level-set equivalent function) are also shown as follows, respectively.

$$\nabla \cdot \mathbf{u}_i = 0 \quad (1)$$

$$\rho \frac{\partial \mathbf{u}_i}{\partial t} + \rho \mathbf{u}_j \frac{\partial \mathbf{u}_i}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} \quad (2)$$

$$\frac{\partial \alpha}{\partial t} + \frac{\partial \alpha \mathbf{u}_i}{\partial x_i} = 0 \quad (3)$$

$$\frac{\partial \phi}{\partial t} + \frac{\partial \phi \mathbf{u}_i}{\partial x_i} = 0 \quad (4)$$

where, u_i is the speed in the i direction, m/s; ρ is fluid density of molten polymers, kg/m³; x is coordinate component, m; x, j is coordinate direction; p is pressure, Pa; τ is strain tensor, Pa; α is volume fraction, 0 $\sim$ 1; ϕ is the distance from the point to the interface, m.

2.3.3. Constitutive model

Bird-Carreau model was selected to fit the viscosity variation curves of the samples with wide shear rates ranges.

$$\frac{\eta - \eta_{\infty}}{\eta_0 - \eta_{\infty}} = [1 + (\lambda C_a \dot{\gamma})^2]^{\frac{n-1}{2}} \quad (5)$$

where, η is shear viscosity of polymers, Pa·s; η_0 is zero shear viscosity of polymers, Pa·s; η_{∞} is infinite shear viscosity of polymers, Pa·s; λC_a is relaxation time, s; n is non-Newtonian index. Furthermore, due to a stronger affinity of CNTs to PBT phases [38], an assumption was proposed that all CNTs were located in the PBT phases in the mixing progress of PP/PBT/CNTs composites. Therefore, by characterizing the viscosity of the PBT/CNTs blends, the content variations of the CNTs during simulation calculation process could be simplified as viscosity variations of the PBT-CNTs phases, where PBT-CNTs blends with CNTs volume fraction of x were represented as PBT- x CNTs. Hence, Fig.S3 shows shear rate-shear viscosity plots of PP, PBT and PBT- x CNTs blends, and Table S1 shows fitting parameters.

2.4. Characterization

2.4.1. Morphological observation

In order to observe the fractured morphology of the samples, the samples with the size of $50 \times 13 \times 1 \text{ mm}^3$ were placed in liquid nitrogen for 30 min and then cryo-fractured. Some of the cryo-fractured specimens were immersed in a xylene solution at 120°C for 4 h and then cleaned the xylene solution and surface impurities with anhydrous ethanol and deionized water to etch PP substrate completely.

After gold spraying treatment, these obtained cryo-fractured sections were observed under a vacuum scanning electron microscope (SEM, JSM-6360LV, Nippon Electron Co., Ltd., Japan) at an accelerated voltage of 3.0 kV. Energy Dispersive Spectrometer (EDS, Horiba 7021-H) was also used to analyze the elements distribution of the cross-sections. In addition, the phase scale and distribution of the phase morphology were analyzed by Image-Pro Plus 6.0 with a minimum of 100 sets of data.

2.4.2. Rheological testing

A flat plate rheometer (Bohlin Gemini 2, Malvern Instruments, UK) and a capillary rheometer (Rosand RH10-D, Malvern Instruments, UK) were used for steady rheological tests of the samples in the shear rates range of 10^{-2} – 10^1 rad/s and 10^1 – 10^3 rad/s , respectively. The obtained all rheological data were fitted as steady rheological curves at the shear rates ranges from 10^{-2} to 10^3 rad/s . The flat plate rheometer was also used for the dynamic rheological tests of the samples with a controlled strain of 3 % and a shear rate range of 10^{-2} – 10^2 rad/s . For the flat plate rheometer, the samples were hot-compressed into a circular disc with a diameter of 25 mm and a thickness of 1 mm. For the capillary rheometer, the outlet diameter of capillary was 1 mm and the capillary aspect ratio was 16. The temperature in all rheological tests was set at 250°C .

2.4.3. Electrical conductivity measurement

Each sample was prepared into the size of $10 \times 10 \times 1.8 \text{ mm}^3$ for electrical conductivity measurement. For the samples with the electrical conductivity higher than 10^{-6} S/m , a four-probe measuring system (ST2253, Suzhou Jingge Electronics Co., Ltd., China) was used, and for the electrical conductivity lower than 10^{-6} S/m , a high resistance meter (PC68, Shanghai Precision Scientific Instrument Co., Ltd., China) was used at the room temperature (25°C). No less than 5 specimens in each group of samples were tested and the average value was recorded.

2.4.4. EMI SE characterization

The EMI SE of the samples in X-band frequency range (8.2–12.4 GHz) was determined using a vector network analyzer (E5071C, Keysight, USA). The samples with the size of $23 \times 10 \times 2 \text{ mm}^3$ were placed in a specimen fixture and tested by waveguide method. The S -parameters

(S_{11} , S_{21} , S_{12} and S_{22}) were recorded and used for calculating the total EMI SE (SE_T), the absorption contribution (SE_A), the reflection contribution (SE_R) according the following equations:

$$SE_R = -10\lg(1 - |S_{11}|^2) \quad (6)$$

$$SE_A = -10\lg\left(\frac{|S_{21}|^2}{1 - |S_{11}|^2}\right) \quad (7)$$

$$SE_T = SE_R + SE_A + SE_M \quad (8)$$

where, S_{11} and S_{21} represent return loss and insert loss parameter; T , R and A represent transmission, reflection and absorption coefficient; SE_M represents multiple reflection shielding effectiveness, dB, which could be negligible when $SE_T \geq 15 \text{ dB}$ [39].

2.4.5. Mechanical property testing

Each sample was made into V-shaped standard specimens according to ASTM D638-14. The gauge length of standard specimens was 20 mm and the thickness was 1.8 mm. A universal tensile testing machine (RGM-2020, Shenzhen Rigel Tensile Tester Co., Ltd., China) was used for tensile testing of the standard specimens with a tensile speed of 5 mm/min at room temperature (25°C). No less than five specimens in each group of samples were tested and recorded for calculating the average value and standard deviation.

3. Results and Discussion

3.1. Construction of co-continuous structure of PP/PBT blends

In order to investigate the construction process of co-continuous structure in PP/PBT blends, the phase morphology and fluid mixing process of PP/PBT blends with volume ratio of 8:2, 5:5 and 2:8, respectively, are shown in Fig. 1 (A) and (B1). When the volume ratio is 80:20, most of the PBT phase was dispersed within the PP matrix. In the independent PP fluid diagram (away PBT), PP droplets were highly interconnected, whereas in the independent PBT fluid diagram (away PP), PBT droplets were only weakly interconnected and mostly isolated. When the volume ratio is 50:50, the PP and PBT could be no longer completely separated, and the interfacial region between them thickened and became more entangled. After either PP or PBT was removed, the remaining droplets in the blends remained highly interconnected. When the volume ratio is 20:80, When the PP/PBT volumetric ratio was 20:80-similar to a volumetric ratio of 80:20-the PBT phase formed a continuous network, while PP droplets were dispersed independently. Fig. 1 (B2) shows the average viscosity variations of PP, PBT and their blends during mixing process for better understanding the formation of co-continuous structure. Fig.S4 and Fig.S5 in Supporting Information show also the formation process of the co-continuous structure in PP/PBT (50:50) blends.

3.2. Selective migration of CNTs in PP/PBT blends

To investigate the selective localization of CNTs in co-continuously structured PP/PBT blends, PP/PBT/CNTs composites were prepared using different mixing sequences. The volume ratio of PP to PBT was fixed at 50:50, and the CNTs content was maintained at 5 vol%. As shown in Fig. 2 (A) that, the PP and PBT phases formed a continuous structure in which one phase exhibited a rougher surface and the other a smoother one, indicating that CNTs selectively localized in one of polymer phase. To determine the specific phase in which CNTs resided within the PP-CNT/PBT composites, energy-dispersive-spectrometer (EDS) were performed, as shown in Fig. 2 (B1) and (B2), respectively. The CNT distribution closely overlapped with the oxygen (O) distribution originating from the PBT molecular chains, indicating that CNTs migrated preferentially from PP to PBT and selectively localized within

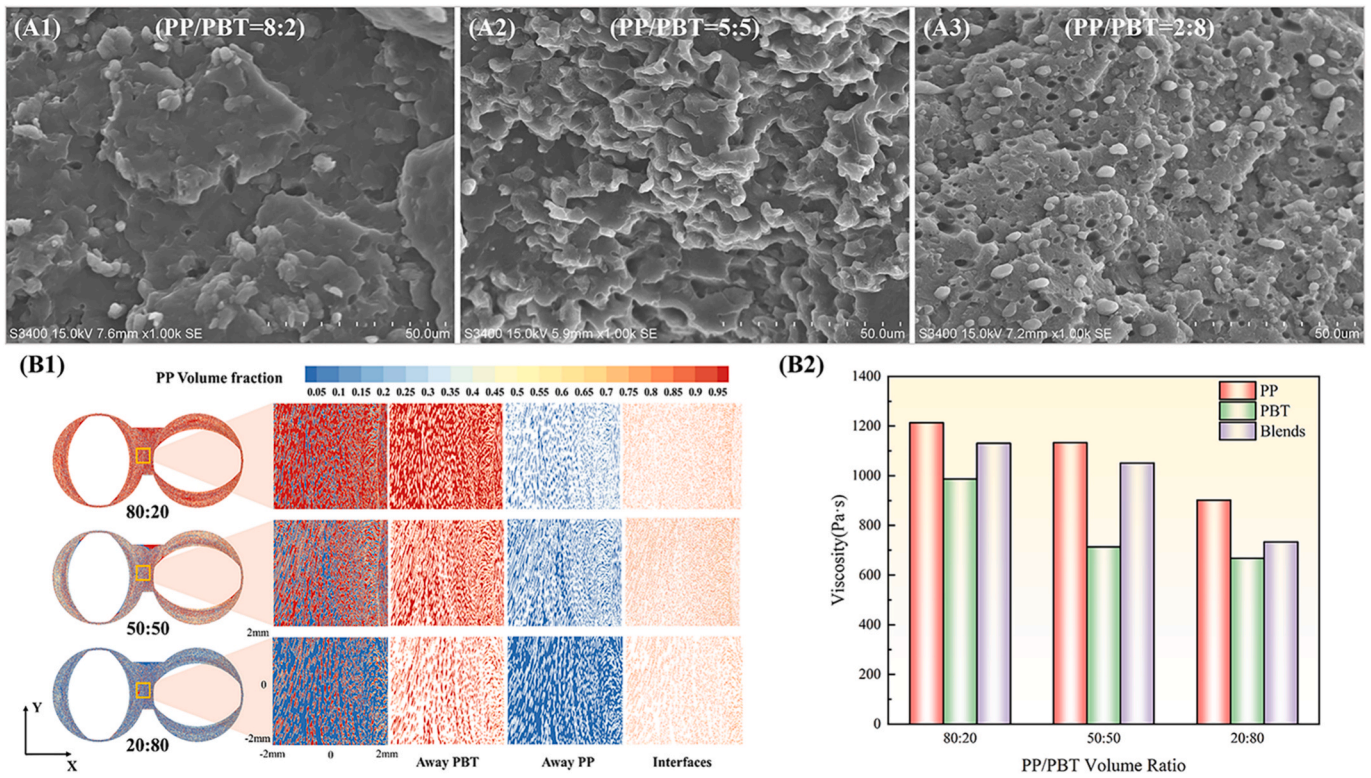


Fig. 1. (A) SEM images, (B) Fluid mixing process and viscosity variation of PP/PBT blends with different volume ratios.

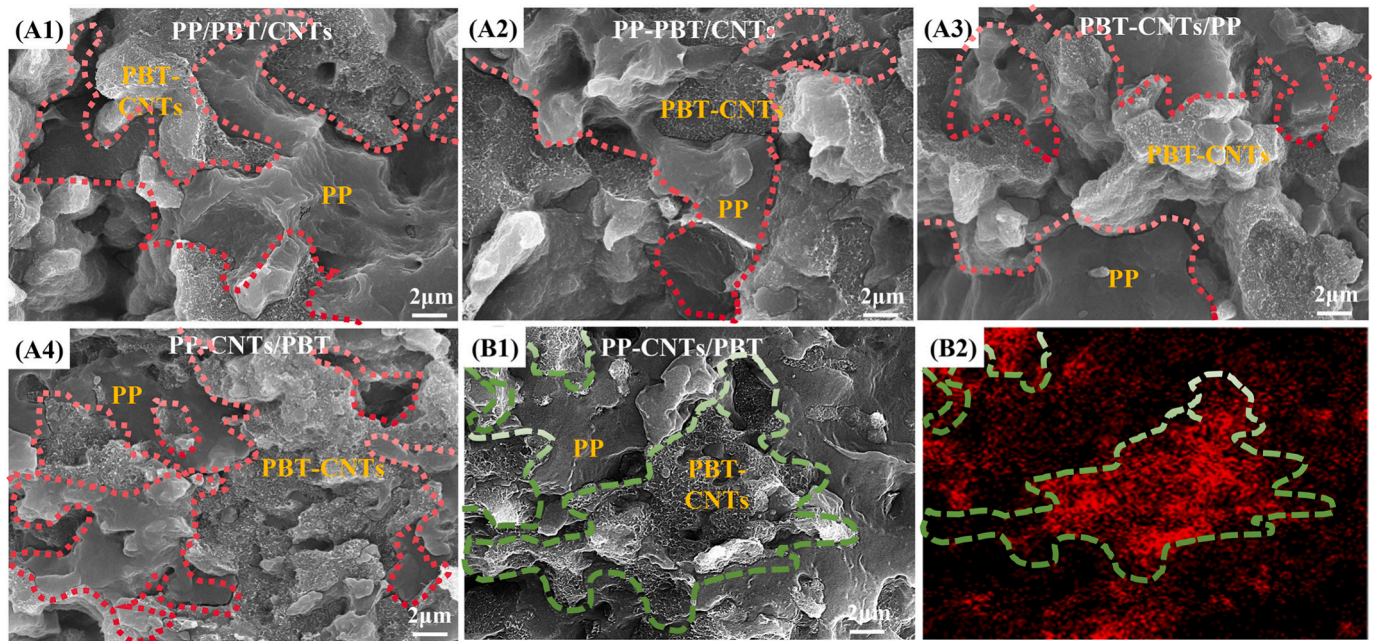


Fig. 2. (A) SEM images of PP/PBT/CNTs composites prepared using different methods. (B) SEM images (B1) and O distribution in PP-CNTs/PBT composites (B2).

the PBT phase during mixing when CNTs were initially blended with PP.

In incompatible polymer systems, nanofillers directionally migrate between different polymers and thermodynamically equilibrate, and nanofillers' selective localization can be theoretically predicted by calculating the wetting coefficients according to Young's equation [40] proposed by Sumita et al., as follows:

$$\omega = \frac{\gamma_{CNTs-PBT} - \gamma_{CNTs-PP}}{\gamma_{PBT-PP}} \quad (9)$$

where, geometric-mean equation [41] was used to calculate the interfacial energy (γ) between each component. Based on the literature [42], the surface tension values of PP, PBT and CNTs at 250 °C are shown in Table S1. Calculations yielded $\omega = -1.25$. Therefore, according to Ref. [40], the CNTs had a stronger affinity for the PBT phases due to thermodynamic effects. This also further supports the SEM-EDS experiment results based on thermodynamic kinetics theory.

3.3. Regulation of the Phase-Domain scale of PP/PBT/CNTs composites

Fig. 3 (A) shows SEM images of PP/PBT/CNTs composites prepared using different blending methods after etching PP phases. The average domain scale of the PBT phases was statistically analyzed (Left Y-Bottom X in Fig. 3 (C)). For the PP/PBT with co-continuous structure, the average PBT phase domain size was as big as 13.8 μm . After the CNTs were added, the average PBT phase domains of the PP/PBT/CNTs, PP-PBT/CNTs, PP-CNTs/PBT, and PBT-CNTs/PP composites were calculated at 2.5, 2.2, 1.8, and 3.1 μm , respectively, indicating that the addition and migration of CNTs had certain effects on refining phase size. To quantitatively analyze the influence of CNTs movement on phase domain size, Palierne model [43] was introduced to this work, which could correlate the viscoelastic behavior of multiphase polymer

systems with phase morphology parameters and interfacial tension.

$$G_{blend}^* = G_m^* \frac{1 + 3\phi H(\omega)}{1 - 2\phi H(\omega)} \quad (10)$$

where, $H(\omega)$ depends on the interfacial tension and phase domain size. Fig. 3 (D) shows storage modulus and viscous modulus of PP/PBT as well as PP/PBT/CNTs composites and their corresponding fitting curves from Palierne model. The obtained interfacial tension of different samples after fitting is shown in Right Y-Bottom X of Fig. 3 (C). The fitting curves can well coincide with the rheological data, indicating that Palierne model closely correlated linear viscoelastic behavior of PP/PBT systems with their phase domain size and interfacial tension [44]. However, Palierne model has also certain limitations, such as being based on small amplitude oscillatory shear theory, which would cause

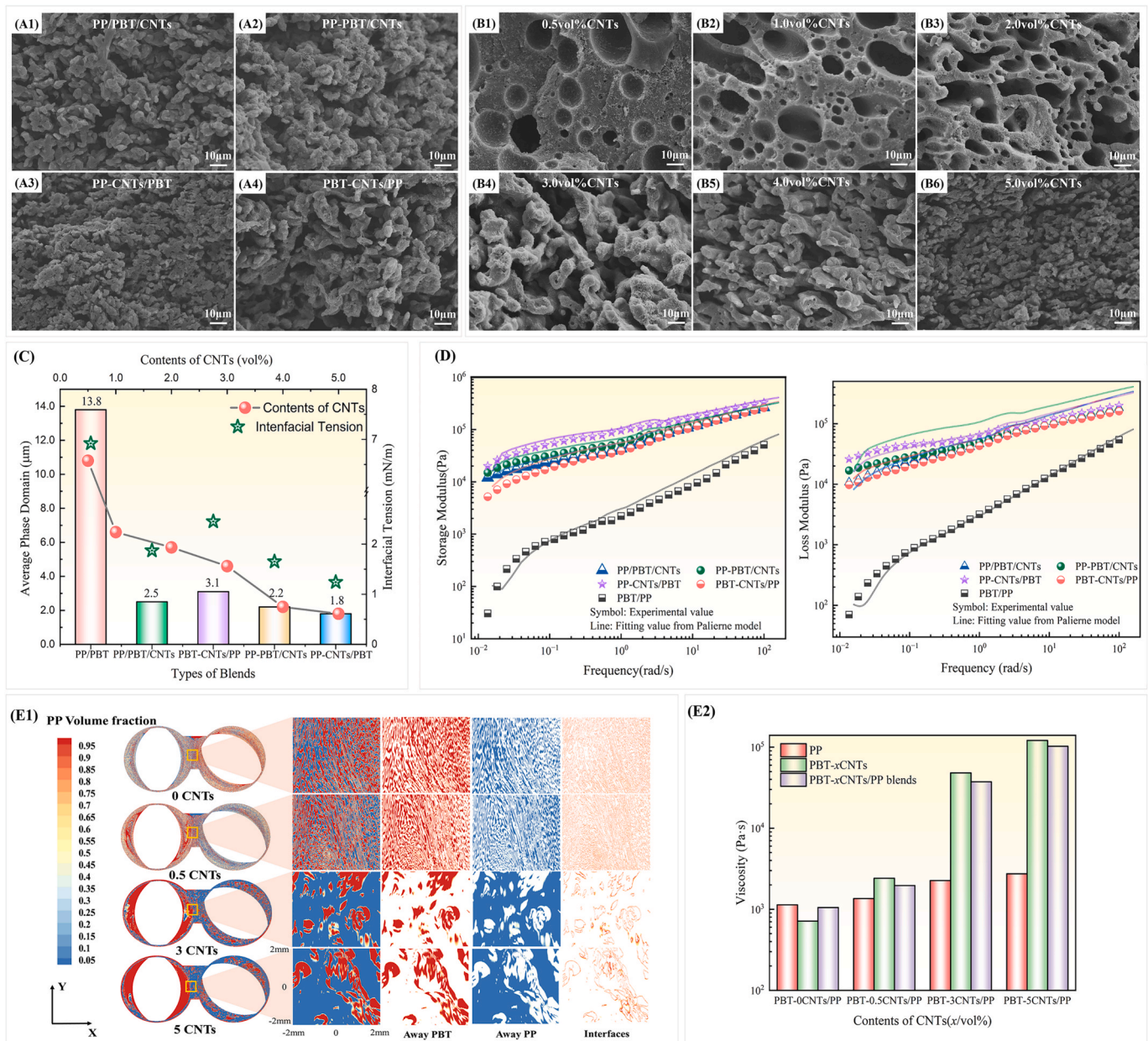


Fig. 3. (A) SEM images of PP/PBT/CNTs composites prepared using different blending methods after etching PP phases (B) SEM images of PP-CNTs/PBT composites prepared using different CNT contents after etching PP phases (C) Average phase domain size of PP/PBT/CNTs composites (Left Y-Bottom X), average phase domain size of PP-CNTs/PBT composites (Left Y-Top X) and interfacial tension of PP/PBT/CNTs composites calculated based on Palierne model (Right Y-Bottom X) (D) Storage modulus and viscous modulus of PP/PBT/CNTs composites and the corresponding fitting curves from Palierne model (E1) Fluid mixing process of PBT-xCNTs/PP composites and (E2) the corresponding viscosity variation bar of polymer phases.

the fitted curves to deviate in the low-frequency range. Nevertheless, the obtained interfacial tension for PP/PBT systems is yet 6.98 mN/m, which is generally consistent with 7.2 mN/m obtained from Ref. [45], further confirming the role of Palierne model. After the addition of CNTs, the fitted interfacial tension values of the composites decline to below 3.0 mN/m, indicating that some CNTs acted as effective compatibilizers at PP-PBT interface, significantly improving compatibility. Moreover, the obtained interfacial tension for PP-CNTs/PBT composites is lowest, 1.24 mN/m, indicating that when CNTs migrated from PP to PBT, parts migrated into the interior of PBT phase, while other parts located at the phase interface. These CNTs at the interface played important roles: forming physical cross-linking at the interface; reducing the interface energy between two phases; inhibiting droplet coalescence, all of which resulted in the refinement of phase domain size. In case of other three composites, it is known from Fig. 2 that less CNTs movement in the mixing process resulted in a lower probability of being located at phase interface, hence, the obtained phase domain size was relatively coarsening.

Fig. 3 (B) shows SEM images of PP-CNTs/PBT composites with different CNTs contents after etching PP phases. As CNTs content increases, a gradual evolution of two-phase morphology from sea-island structure to co-continuous structure could be observed. And the average phase domain size of PBT phases decreases from 13.8 μm to 1.8 μm , indicating that an increase of CNTs' content was also an important influencing factor for refining phase scale in composites.

To further clarify the CNTs' migration process, fluid mixing process of PBT-xCNTs and PP calculated using VOF-Level-set simulation coupling method is shown in Fig. 3 (E1), and Fig. 3 (E2) is corresponding viscosity variation bar of polymer phases. It should be noted that the simulation process neglected the migration of CNTs from PP to PBT, as mentioned in Section 2.3.3. Clearly, when the CNTs' contents in the composites were 0 and 0.5 vol%, the viscosity ratios λ of the PBT-xCNT phase to the PP phase were 0.63 and 1.78, respectively, converging to 1, while phase separation and dispersion accelerated, and the two-phase fluids mutually entangled, forming complex and rich interfaces. After either PP or PBT was removed, the remaining phase showed good continuity. When the CNTs contents were 3 and 5 vol%, the viscosity ratios λ of the PBT-xCNT phase to the PP phase were as high as 21.29 and 44.18, respectively. Additionally, phase dispersion and mixing slowed, hindering the division of the PBT phase by the lower-viscosity PP phase into a smaller phase-domain structure, thereby substantially decreasing in the interfacial area between the phases. According to Taylor's theory and Lee-Park's model [46,47], after the fragmentation and coalescence of droplets reaching dynamic equilibrium, steady-state droplet dimension d can be expressed as:

$$d = \frac{C \cdot \sigma}{\eta_c \cdot \dot{\gamma}} Ca_{crit}(\lambda) \quad (11)$$

where $Ca_{crit}(\lambda)$ is function of viscosity ratio λ of two phases. When the viscosity ratio λ approaches 1.0, the critical capillary number Ca_{crit} is minimized, and dispersed phase droplets are most easily fragmented into smaller sizes by shear fields [48,49]. Therefore, according to the simulation results, it can be concluded that, when CNTs content was 0.5 vol% or 1.0 vol%, the viscosity ratio λ window essentially guided the PP/PBT system towards the most favorable phase refinement predicted by Taylor's theory. However, especially for PP-CNTs/PBT composites, the influence of thermodynamic CNTs migration and interfacial tension on the refinement of phase domain size should be emphasized.

In this work, when the blending sequence of PP-CNTs/PBT was adopted, the migration of CNTs from PP to PBT phase dynamically adjusted the viscosity of two phases. In the initial stage, PP-CNTs masterbatch had high viscosity, while PBT had lower viscosity, as shown in Fig. S3. At this moment, the viscosity ratio λ_{PBT-PP} was smaller. As CNTs migrated, the viscosity of PBT phase increased with the addition of fillers, while the viscosity of PP phase decreased with the decrease of fillers, resulting in the λ_{PBT-PP} gradually approaching 1. This trend of

approaching-approximation-increasing of λ_{PBT-PP} directed PP/PBT towards the viscosity ratio window predicted by Taylor's theory [34], which is most favorable for phase refinement. Moreover, the migration process of CNTs inevitably passed through PP-PBT interface, and the CNTs at the phase interface intensified the frictional partitioning behavior during the two-phase mixing process, which also played a certain role in capacity expansion, reduced the phase interface energy and improved the microstructure stability [50]. Therefore, we observed a significant refinement in PBT phase domain size from 10.8 μm to 1.8 μm as CNTs content increased from 0.5 vol% to 5 vol%.

Based on these results, a schematic of the proposed mechanism by which CNTs migration regulates the phase scale of the PP/PBT blends is shown in Fig. 4. A comparison revealed that when the PBT-CNTs masterbatch method was used to prepare the composites, the viscosity ratios of the PP and PBT phases negligibly changed because the CNTs were predominantly in the PBT phase, which continuously enlarged the phase-domain scale. Meanwhile, the CNTs were only subjected to the external flow field, completing the dispersion and increasing the agglomeration. Nevertheless, when the PP-CNTs masterbatch method was used, the viscosity ratios of the PP and PBT phases converged and increased because CNTs migrated from PP to PBT, promoting frictional division between the phases and continuously refining the phase-domain scale of the co-continuous structure. Moreover, under the combined influence of their own migration and the external flow field, the CNTs in the finer polymer-phase domain were subjected to more intense volumetric repulsion, improving CNTs dispersion. In addition, numerous complex and intertwined polymer interfaces and denser conductive pathways were constructed by the refined phase-domain structure containing more evenly dispersed CNTs. Theoretically, such structures strengthen the ability of composites to reflect electromagnetic waves while enhancing the absorption loss of waves that have penetrated the material [51–53].

3.4. Electrical Conductivities, EMI-SE and mechanical properties of PP/PBT/CNTs composites

Fig. 5 (A) shows electrical conductivity plotted as functions of the CNTs content for the PP/PBT/CNTs composites prepared using different blending methods. When the CNTs content increases from 0 to 0.5 vol%, the electrical conductivities of all the samples transitions completely from high to low resistance. This transition was attributed to the strong volumetric repulsion of the selectively distributed CNTs within the co-continuous structures [54,55], which facilitated the formation of a continuous conductive network through the long-range connectivity of CNT fillers in the composites. Compared with the composites prepared using the other blending methods, the PP-CNTs/PBT composites exhibited higher electrical conductivities at the same CNTs contents. This was because the smaller PBT domain scale in the PP-CNTs/PBT composites exerted a stronger volumetric repulsion effect on the CNTs [56], which reduced CNTs agglomeration and promoted the formation of a denser, more homogeneous conductive network within the finer phase structures; On the other hand, during the preparation of the PP-CNTs/PBT composites, the CNTs were dispersed more uniformly owing to their directional migration, resulting in the formation of a more well-developed conductive network structure.

Fig. 5 (C) shows the EMI-SE of the PP/PBT/CNTs composites prepared using different blending methods and CNT contents. As shown in Fig. 5 (C1), for the composites prepared using all the blending methods, EMI-SE_T remained stable with changing electromagnetic wave frequency in the X-band, with average values of 32.6, 34.1, 35.6, and 27.1 dB for PP/PBT/CNTs, PP-PBT/CNTs, PP-CNTs/PBT, and PBT-CNTs/PP, respectively. Moreover, the PP-CNT/PBT composites exhibited higher reflection and absorption SE, reaching 29.7 and 5.94 dB, respectively, mainly because of the smaller phase-domain scale and denser conductive network structure in the PP-CNT/PBT composites, as shown in the diagram of the proposed mechanism of the electromagnetic wave

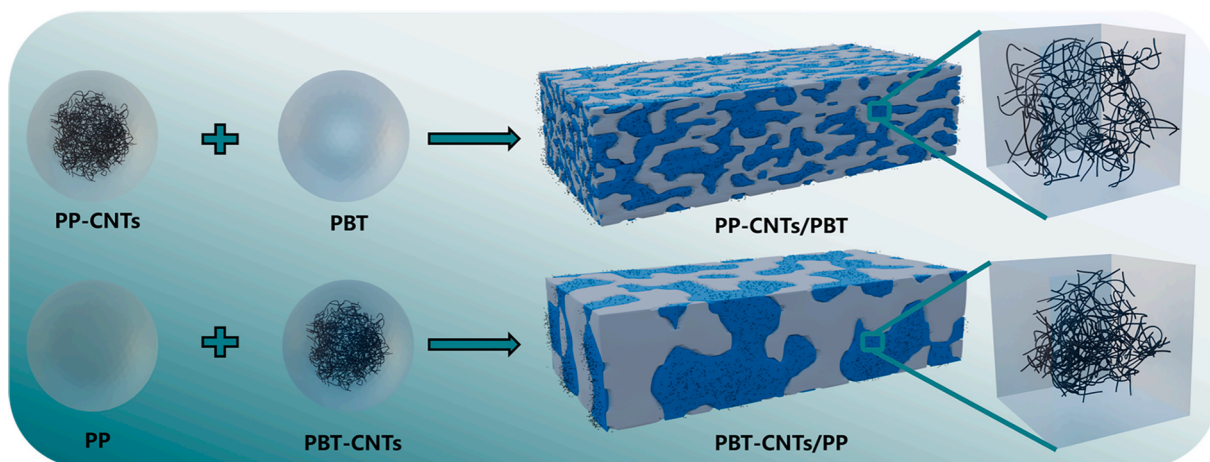


Fig. 4. Schematic of proposed mechanism of migration-induced CNTs dispersion and phase-domain refinement.

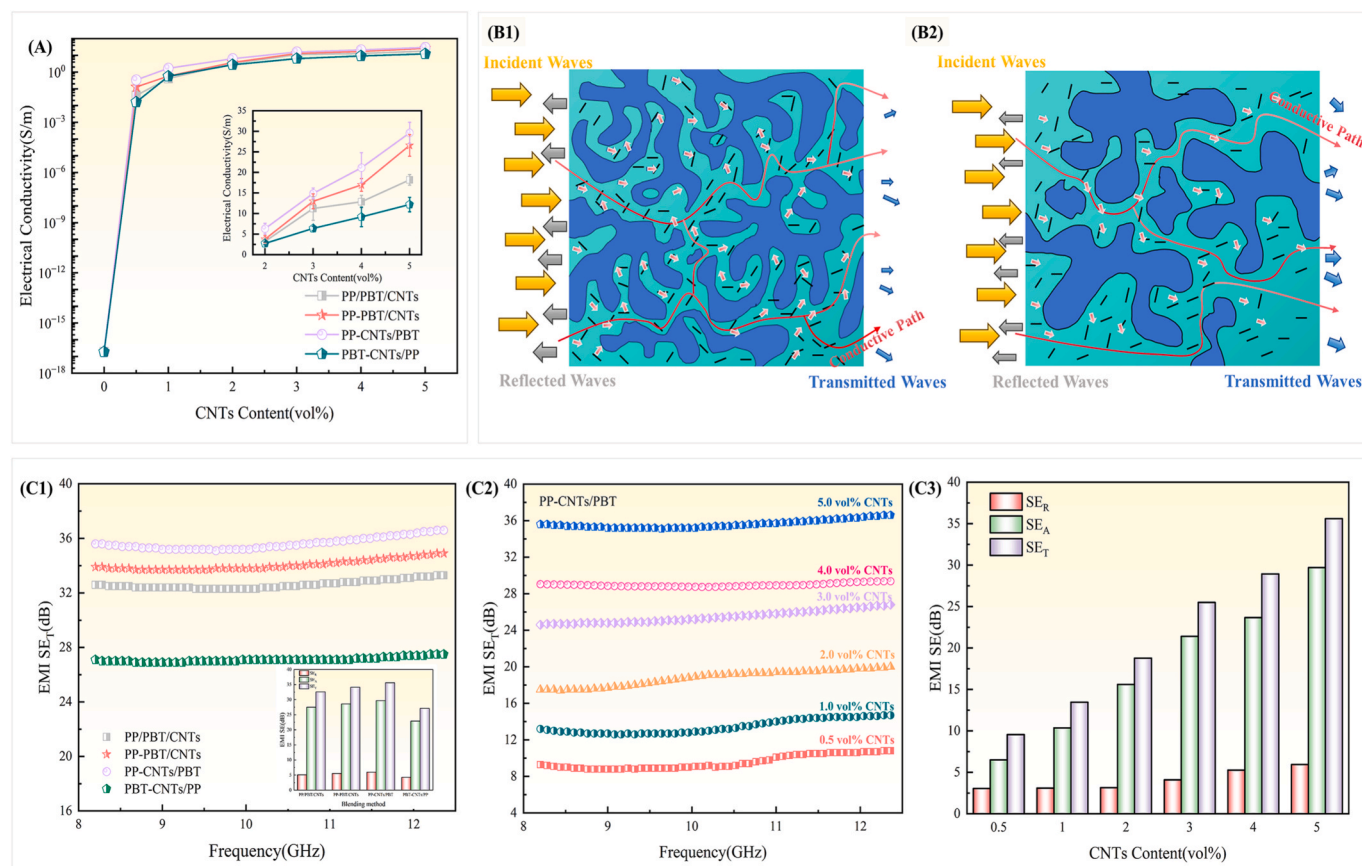


Fig. 5. (A) Electrical conductivity plotted as functions of the CNTs content for the PP/PBT/CNTs composites prepared using different blending methods. (B) Diagram of the proposed mechanism of electromagnetic wave transmission in (B1) PP-CNTs/PBT and (B2) PBT-CNTs/PP composites. (C) EMI-SE plotted as functions of frequency for the (C1) PP/PBT/CNTs composites prepared using different blending methods and (C2, C3) PP-CNTs/PBT composites prepared using different CNT contents. All the samples were 2 mm thick.

transmission (Fig. 5 (B)). Compared with the PBT-CNTs/PP composites, the PP-CNTs/PBT composites exhibited a refined phase morphology. The formation of the co-continuous structure generated richer and more complex polymer-phase interfaces and a denser, more efficient conductive network, which enhanced internal electron transport pathways and efficiency, thereby improving both the electrical conductivity of the composites and the reflection efficiency of electromagnetic waves on their surfaces. Moreover, the increased internal area and phase boundaries of the composites also contributed to multiple reflection and

absorption losses of electromagnetic waves, improving the absorption efficiency and, thus, optimizing the EMI-SE_T of the composites.

Fig. 5 (C2) shows EMI-SE plotted as functions of frequency for the PP-CNTs/PBT composites prepared using different CNTs contents. The overall EMI-SE of the composites increased progressively with increasing CNTs content. Furthermore, when the CNTs content reached 3 vol%, the EMI-SE of the as-prepared PP-CNTs/PBT composites met the demands of commercial electromagnetic shielding materials (> 20 dB). In addition, as shown in Fig. 5 (C3), with increasing CNTs content, both

SE_A and SE_R gradually increased for the composites. The average reflection coefficient (R) corresponding to each CNTs content was 0.50, 0.51, 0.52, 0.61, 0.70, and 0.75, indicating that reflection shielding was the main mechanism by which the composites shielded electromagnetic waves. After penetrating the composites, electromagnetic waves underwent multiple reflection losses because of the gradually refined phase-domain structure, and the absorption efficiency continued to increase, while the transmission coefficient was gradually reduced.

Fig. 6 shows the tensile properties of the PP/PBT/CNT composites prepared using different CNT contents and blending methods. Clearly, all the PP/PBT/CNT composites exhibited brittle fracture. When the CNT content was increased from 0 to 0.5 vol%, both the tensile strength and elongation at break substantially improved for the composites prepared using the different blending methods, and the most obvious improvement is observed for PP-CNTs/PBT composites, from 16.4 MPa and 38.1 % to 27.8 MPa and 42.1 %. Then, the tensile strength and elongation at break of the composites show rapid decrease until CNTs' content reaches 2 vol%. Subsequently, both the tensile strength and elongation at break rapidly decreased until the CNT content reached 2 vol%. Subsequently, the mechanical properties of the composites, especially the PP-CNT/PBT composites, remained relatively stable. After adding CNTs into PP/PBT, the improvement of mechanical property of the composites was due to mechanical properties of nano-fillers themselves. When the CNTs content was increased from 0.5 to 2 vol%, the phase viscosity and domain scale of the composites constantly changed because CNTs selectively migrated, while the considerable added nanofiller content might have concentrated stress within the polymer composites, thus affecting the overall mechanical properties. When the CNT content reached 3 vol%, the tensile strength of the PP-CNTs/PBT composites slightly decreased, while the corresponding elongation at break slightly increased because in PBT phases, the CNTs content saturated, and CNTs gradually migrated to the interface. The refined phase domains and construction of the complex interfaces of co-continuous structures was helpful for transferring stress in the composites, reducing stress concentration, and, thus, resisting the negative effects of the nanofiller-induced mechanical property degradation.

Table 1 compares the percolation thresholds, EMI-SE, tensile strength, and elongations at break of several polymer/carbon-based composites reported in the literature with those of the PP/PBT/CNTs composites prepared in this work. Evidently, at lower filler contents, the as-prepared PP-CNTs/PBT composites possessed an ultralow percolation threshold and a higher EMI-SE while maintaining superior mechanical properties for practical applications.

3.5. Impacts of compatibilizers on PP-CNTs/PBT composites towards EMI SE

The above results showed that the refined phase morphology can improve the EMI-SE of the composites. Additionally, because suitable compatibilizers can further regulate the polymer-phase morphology of

the co-continuous structure [64–66], PP-3CNTs/PBT composites, which satisfied the EMI-SE requirements of commercial EMI materials, were blended with different compatibilizers for further investigating the effects of phase-scale regulation on EMI-SE.

Fig. 7 (A)–(D) shows the SEM images of the PP/PBT composites blended with different compatibilizers. Clearly, in the PP/PBT composites, PP phases were spherical and uniformly distributed in continuous PBT phases, with an average phase domain of 16.1 μm . After different compatibilizers were added, the average PP phase domain was substantially smaller because the compatibilizers effectively reduced the interfacial tension and enhanced the shear between PP and PBT, substantially shrinking the phase domain [65]. Furthermore, the PP-g-GMA compatibilizer forced the polymer phases to form smaller domains because the GMA group reacted with PBT molecular chains, enhancing the shear fragmentation of both phases.

Fig. 7 (E1)–(H1) shows the SEM images of the PP-CNTs/PBT composites blended with different compatibilizers, and Fig. 7 (E2)–(H2) shows the corresponding SEM images of the same blended composites after the PP phases were etched, where the CNT and compatibilizer contents were 3 vol% and 5 parts per hundred rubber (phr), respectively. Clearly, the cryo-fractured PP-CNTs/PBT composites surfaces contained many pores, and obvious gaps had formed at the interface between the two phases, indicating poor compatibility between PP and PBT. After the compatibilizers were added, the gaps narrowed between PP and PBT in all the samples, indicating that all the tested compatibilizers helped to improve the compatibility between PP and PBT. Moreover, the PBT domain size in PP-CNTs/PBT composites with PP-g-GMA shows smallest value, 3.3 μm , indicating that PP-g-GMA induced a stronger phase-domain refinement, likely because it reduced the interfacial tension between PP and PBT, thereby continuously refining the phase domains during CNT migration. However, PP-g-MAH and POE-g-GMA only interfaced with one of the two phases, easily forming the core-shell structure with the connected polymer phase [67], thereby weakening phase fragmentation during CNT migration.

Fig. 8 (A1) presents the EMI-SE of the PP-CNTs/PBT composites blended with different compatibilizers. The insets show the bar charts for the absorption (SE_A), reflection (SE_R), and total shielding efficiency (SE_T). Clearly, the compatibilizer-free composites exhibited an average SE_T of 25.5 dB. When PP-g-MAH, PP-g-GMA, and POE-g-GMA were incorporated into the composites, the SE_T values reached 25.0, 27.1, and 26.2 dB, respectively. This variation trend matched that of the polymer domain size, further supporting the relationship between the polymers' phase-domain size and composites' EMI. Fig. 8 (A2) presents the EMI-SE of the PP-CNTs/PBT composites blended with different PP-g-GMA contents, and the SEM images of the corresponding samples after the PP phases were etched are shown in Fig. S5. As shown in Fig. 8 (A2), the EMI-SE of the composites remained relatively stable with increasing frequency. And as the content of PP-g-GMA increases, the EMI SE of PP-CNTs/PBT composites exhibit an increase trend until the content reaches 5 phr, indicating that a compatibilizer content of 5 phr was the

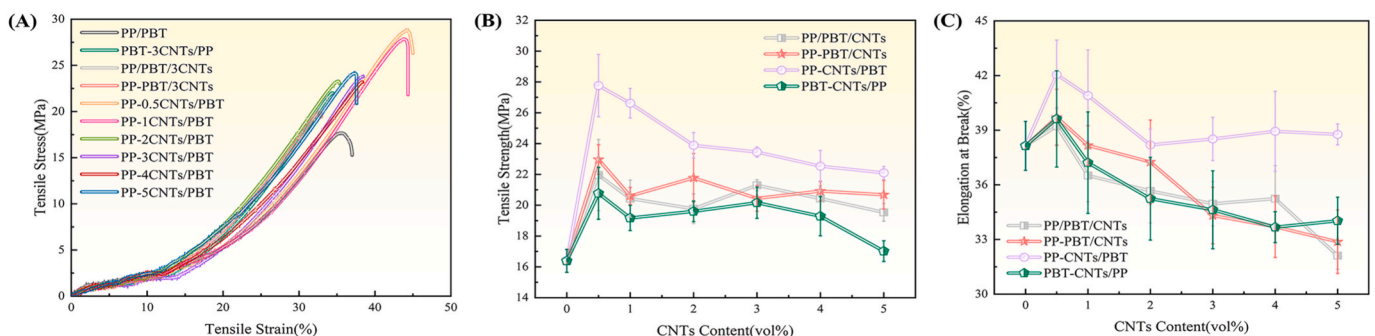


Fig. 6. (A) Stress–strain curves of PP/PBT/CNTs composites prepared using different blending methods and CNTs contents. (B) Tensile strength– and (C) elongation at break–CNTs content curves of PP/PBT/CNTs composites prepared using different blending methods.

Table 1

Comparison of percolation thresholds, EMI-SE, tensile strength, and elongations at break of several polymer/carbon-based composites reported in the literature with those of the PP/PBT/CNTs composites prepared in this work.

Polymer Composites	Filler Content	Thickness	Percolation Threshold	EMI-SE	Tensile Strength	Elongation at Break	Ref.
SEBS/CB	15 wt%	5 mm	5 wt%	17.6 dB	N/A	N/A	[57]
PBT/PET/MWCNTs	10 wt%	4 mm	N/A	7.2 dB	42.8 MPa	N/A	[58]
PBT/PET/GNPs	10 vol%	4 mm	N/A	27.3 dB	53.0 MPa	N/A	[58]
PU/LM	40 vol%	0.3 mm	N/A	34.1 dB	1.1 MPa	1281.8 %	[13]
porous PA6/CB	50 wt%	10 mm	20 wt%	37.3 dB	1.9 MPa	N/A	[59]
segregated POE/CNTs	2.10 vol%	2 mm	0.20 vol%	34.7 dB	3.3 MPa	500 %	[60]
layered TPU/CNTs	6 wt%	1.6 mm	N/A	27.1 dB	N/A	N/A	[61]
PBT/CFs foams	30 wt%	10 mm	N/A	10 dB	N/A	N/A	[38]
PS/PMMA/MWCNTs foams	1.61 vol%	2 mm	0.14 vol%	23.1 dB	15.6 MPa	9.8 %	[62]
co-continuous PLA/PCL/SiO ₂ /MWCNTs	7 wt%	2 mm	0.06 vol%	26 dB	63 MPa	7 %	[23]
co-continuous EMA/EOC/MWCNTs/CB	20 wt%	1 mm	N/A	34.7 dB	8.1 MPa	333 %	[63]
co-continuous PP/PBT/CNTs	5 vol%	2 mm	0.19 vol%	35.6 dB	23.1 MPa	38.9 %	This work
co-continuous PP/PBT/PP-g-GMA/CNTs	3 vol%	2 mm	0.14 vol%	27.6 dB	30.2 MPa	51.8 %	This work

N/A: not available.

SEBS: styrene ethylene butylene styrene; PET: polyethylene terephthalate; GNPs: graphene nanoplates; PU: polyurethane; LM: liquid metal; PA6: polyamide 6; POE: polyolefin elastomer; TPU: thermoplastic polyurethane; CFs: carbon fibers; PS: polystyrene; PMMA: poly(methyl methacrylate); PLA: polylactic acid; PCL: polycaprolactone; EMA: ethylene-methyl acrylate copolymer.

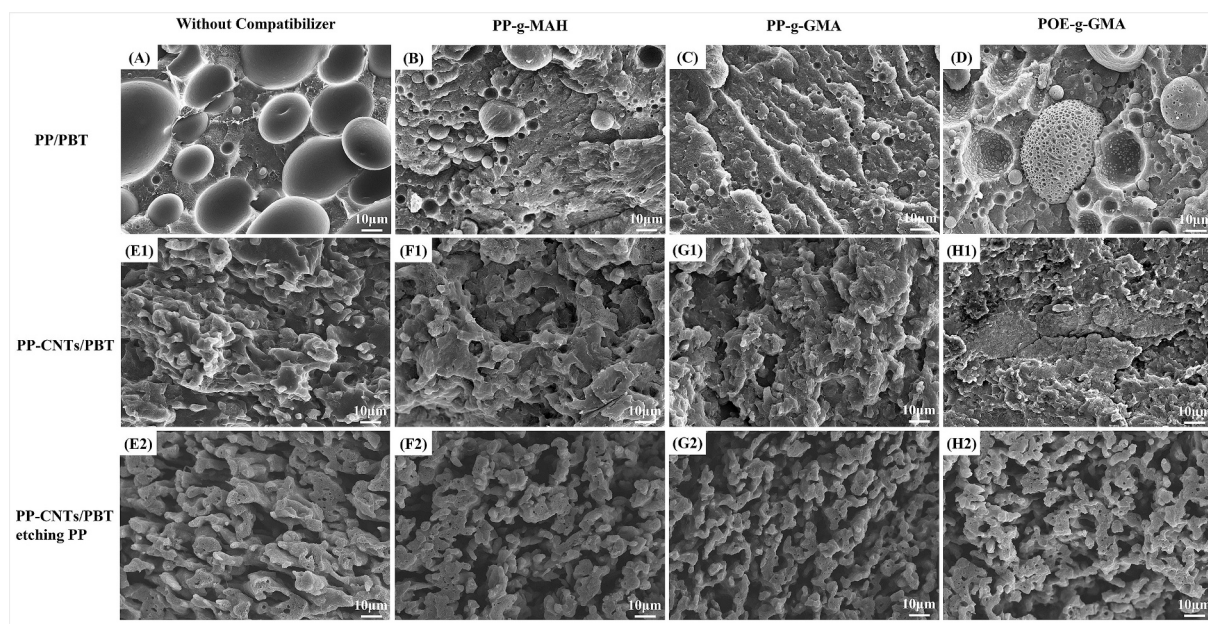


Fig. 7. SEM images of the (A–D) PP/PBT and (E1–H1) PP-CNTs/PBT composites blended with different compatibilizers and the (E2–H2) PP-CNTs/PBT composites blended with different compatibilizers after the PP phases were etched.

threshold for increasing the composites' EMI-SE.

Fig. 8 (B) presents the tensile properties of the PP-CNTs/PBT composites blended with increasing compatibilizer contents. As shown in Fig. 8 (B1), the tensile strength of the compatibilizer-free PP-CNTs/PBT composites was only 23.5 MPa. When 3 phr of either PP-g-MAH or POE-g-GMA was added, the tensile strength increased to 26.8 or 27.8 MPa, respectively. Further increasing the compatibilizer content negligibly enhanced the mechanical properties. When 7 phr of the PP-g-GMA compatibilizer was added, the tensile strength of the composites increased to 31.0 MPa because at higher PP-g-GMA contents, the content of the PP-g-PBT reactant gradually increased, effectively enhancing the interfacial compatibility between PP and PBT. Fig. 8 (B2) reveals the elongations at break of the PP-CNTs/PBT composites blended with different compatibilizers. At compatibilizer contents of 10 phr, the corresponding elongations at break of the PP-g-MAH-, PP-g-GMA-, and POE-g-GMA-modified composites were 44.5, 51.5, and 50.3 %, increasing by 15.6, 33.8, and 30.5 %, respectively, compared with the elongation at break of the compatibilizer-free composite. The superior

enhancements achieved using PP-g-GMA and POE-g-GMA were attributed to the unique characteristics of GMA groups, which reacted with PBT molecular chains. The generated reactants localized at the phase interface, effectively reducing the interfacial tension between PP and PBT. This improved compatibility enabled efficient stress transfer, prevented stress concentration within the composites, and suppressed internal crack initiation, ultimately substantially improving the elongation at break.

4. Conclusion

In PP/PBT/CNTs composites, the phase domain was regulated by controlling CNTs migration, substantially impacting both the conductive threshold and EMI-SE. The PP/PBT/CNTs composites were prepared using different blending sequences and CNTs contents, and CNTs migration was systematically analyzed using both experimental characterization and numerical simulation methods. T CNTs migrated from the PP phase to the PBT phase because of the higher affinity of CNTs for

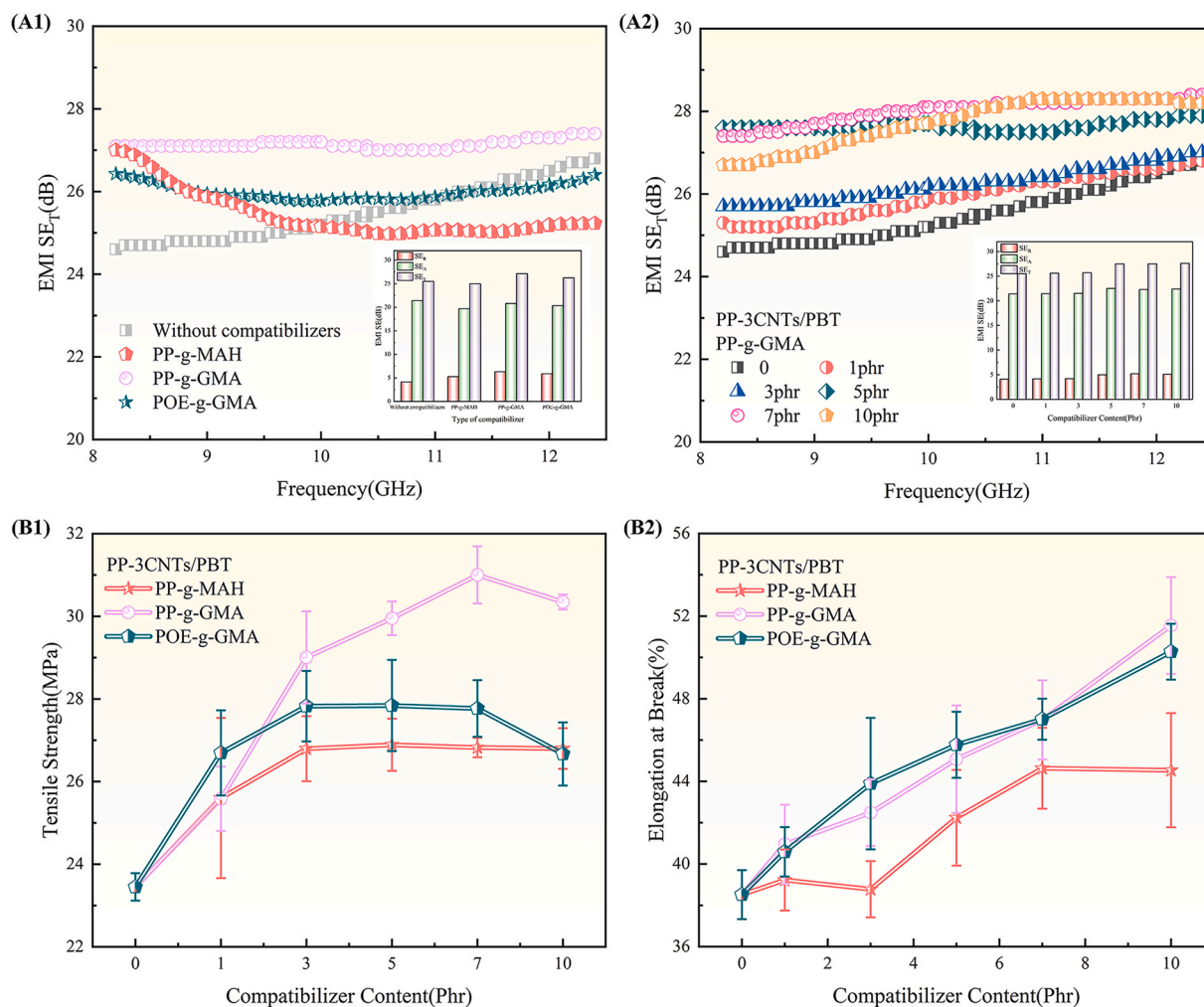


Fig. 8. (A) EMI-SE plotted as functions of the frequency for and (B) mechanical properties of the PP-CNT/PBT composites blended with different compatibilizer types and contents. All the samples for EMI-SE were 2 mm thick.

the PBT phase, even when CNTs were pre-mixed with PP. During the preparation of the PP-CNT/PBT composites, the phase viscosities of both components converged and increased, the phase-domain scale was refined from 3.1 to 1.8 μm , and both the CNT dispersion and compatibility between the phases improved because of CNT migration. Moreover, in the co-continuous structure, the similar viscosities of the component phases contributed to the mutual frictional division of each phase, generating a finer phase domain. At CNTs loadings of up to 3 vol %, CNTs migration to the PBT phase saturated, while the EMI shielding of the composites gradually stabilized because of the construction of double-percolation structures and more complex interfaces in the PP-CNTs/PBT composites, which, at a 5 vol% CNTs loading, exhibited an ultralow conductive percolation of ~ 0.19 vol% CNTs, an excellent EMI-SE of ~ 36 dB, high reflection (a reflection coefficient of 0.75) and absorption losses, a tensile strength of 23.5 MPa, and an elongation at break of 39.1 %. The as-prepared composites satisfied the requirements of commercial electromagnetic shielding materials. Furthermore, by incorporating different compatibilizers, PP-g-GMA reduced the phase-domain size and enhanced the interfacial compatibility between PP and PBT. This not only verified the relationship between phase-domain regulation and EMI shielding in the PP/PBT/CNTs composites but also synergistically improved both the EMI-SE and mechanical properties of the PP-3CNTs/PBT composites, yielding an average SET of 27.1 dB, a tensile strength of 31.0 MPa, and an elongation at break of 51.5 %.

CRediT authorship contribution statement

Yu Wang: Writing – review & editing, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xu Zhang:** Writing – original draft, Investigation. **Guo Li:** Visualization, Supervision, Project administration, Conceptualization. **HuiHao Zhu:** Software, Methodology, Formal analysis. **YuLu Ma:** Supervision, Funding acquisition. **LinSheng Xie:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matdes.2025.115131>.

Data availability

Data will be made available on request.

References

- [1] Z. Cui, M. Yang, G. Han, H. Zhang, Y. Wang, Y. Zhang, Z. Li, J. He, R. Yu, J. Shui, X. Liu, Recent advances in carbon composite films for high-performance, multifunctional and intelligent electromagnetic interference shielding and electromagnetic wave absorption, *Carbon*. 230 (2024) 119627, <https://doi.org/10.1016/j.carbon.2024.119627>.
- [2] Y. Wu, S. Tan, G. Fang, Y. Zhang, G. Ji, Manipulating CNT films with atomic precision for absorption effectiveness-enhanced electromagnetic interference shielding and adaptive infrared camouflage, *Adv Funct Mater.* (2024) 1, <https://doi.org/10.1002/adfm.202402193>.
- [3] X. Tang, Y. Lu, S. Li, M. Zhu, Z. Wang, Y. Li, Z. Hu, P. Zheng, Z. Wang, T. Liu, Hierarchical polyimide nonwoven fabric with ultralow-reflectivity electromagnetic interference shielding and high-temperature resistant infrared stealth performance, *Nano-Micro Lett.* 17 (2025) 82, <https://doi.org/10.1007/s40820-024-01590-3>.
- [4] F.W. Huang, Q.C. Yang, L.C. Jia, D.X. Yan, Z.M. Li, Aramid nanofiber assisted preparation of self-standing liquid metal-based films for ultrahigh electromagnetic interference shielding, *Chem Eng J.* 426 (2021) 131288, <https://doi.org/10.1016/j.cej.2021.131288>.
- [5] Y. Guo, X. Chen, C. Wei, Y. Luo, J. Chen, Y. Zhu, Flexible conductive polymer composite film with sandwich-like structure for ultra-efficient and high-stability electromagnetic interference shielding, *Compos Sci Technol.* 255 (2024) 110717, <https://doi.org/10.1016/j.compscitech.2024.110717>.
- [6] T.T.L. Nguyen, S.J. Cho, J. Ko, D.C.T. Nguyen, M.W. Kim, N.D. Kim, D.S. Lee, Y. Joo, Exceptional electromagnetic interference shielding using single-walled carbon nanotube/conductive polymer composites films with ultrathin, lightweight properties, *Carbon*. 230 (2024) 119567, <https://doi.org/10.1016/j.carbon.2024.119567>.
- [7] Y. Qi, C. Dai, J. Gao, B. Gou, S. Bi, P. Yu, X. Chen, Reinforcement effects of graphite fluoride on breakdown voltage rating and pyro-resistive properties of carbon black/poly (vinylidene fluoride) composites, *Compos A Appl Sci Manuf.* 177 (2024) 107947, <https://doi.org/10.1016/j.compositesa.2023.107947>.
- [8] J. Lin, S. Hou, Z.X. Wang, Y.F. Lin, Y.Z. Liu, Q.Y. Duan, R.P. Nie, D.X. Yan, L.C. Jia, Z.M. Li, Effect of low-density polyethylene on properties of ethylene-vinyl based semi-conductive shielding materials, *Compos Sci Technol.* 262 (2025) 111046, <https://doi.org/10.1016/j.compscitech.2025.111046>.
- [9] Q.S. Liang, Y.F. Lin, H.D. Huang, L.C. Jia, Highly conductive silicone rubber composites enabled by constructing dual-carbon black networks, *Mater Lett.* 401 (2025) 139258, <https://doi.org/10.1016/j.matlet.2025.139258>.
- [10] P. Dubey, M. Gupta, S.I. Kundalwal, Conductive polymer nanocomposite incorporated with carbon nanotubes for effective electromagnetic interference shielding: A numerical study, *Polym Compos.* 45 (2024) 3576–3590, <https://doi.org/10.1002/pc.28011>.
- [11] H.Y. Zhao, M.Y. Yu, J. Liu, X. Li, P. Min, Z.Z. Yu, Efficient preconstruction of three-dimensional graphene networks for thermally conductive polymer composites, *Nano-Micro Lett.* 14 (2022) 129, <https://doi.org/10.1007/s40820-022-00878-6>.
- [12] Y. Ouyang, L. Bai, H. Tian, X. Li, F. Yuan, Recent progress of thermal conductive polymer composites: Al₂O₃ fillers, properties and applications, *Compos A Appl Sci Manuf.* 152 (2022) 106685, <https://doi.org/10.1016/j.compositesa.2021.106685>.
- [13] R.Y. Ma, S.Q. Yi, Z.X. Wang, Y.F. Lin, R.P. Nie, D.X. Yan, L.C. Jia, Highly stretchable liquid metal/poly(urethane-urea) composite with self-healing ability for electromagnetic interference shielding, *Compos B Eng.* 306 (2025) 112808, <https://doi.org/10.1016/j.compositesb.2025.112808>.
- [14] L.C. Jia, Y.F. Yue, J.F. Zeng, Z.X. Wang, R.P. Nie, L. Xu, D.X. Yan, Z.M. Li, A review on thermal and electrical behaviours of liquid metal-based polymer composites, *J. Mater. Chem. C*. 11 (2023) 12807–12827, <https://doi.org/10.1039/d3tc02560h>.
- [15] C.Y. Wang, Y. Wang, F.X. Zou, B.W. Fang, J. Zhao, H. Zhang, J. Guo, L.C. Jia, D. X. Yan, Construction of lightweight, high-energy absorption 3D-printed scaffold for electromagnetic interference shielding with low reflection, *Compos B Eng.* 291 (2025) 112043, <https://doi.org/10.1016/j.compositesb.2024.112043>.
- [16] Y.K. Li, W.J. Li, Z.X. Wang, P.Y. Du, L. Xu, L.C. Jia, D.X. Yan, High-efficiency electromagnetic interference shielding and thermal management of high-graphene nanoplate-loaded composites enabled by polymer-infiltrated technique, *Carbon*. 211 (2023) 118096, <https://doi.org/10.1016/j.carbon.2023.118096>.
- [17] L. He, Y. Chen, X. Shao, Q. Yao, D. Feng, L. Yin, W. Wang, A facile method in fabricating flexible conductive composites with large-size segregated structures for electromagnetic interference shielding, *Macromol Rapid Commun.* (2024) 2400585, <https://doi.org/10.1002/marc.202400585>.
- [18] Y.Y. Wang, J. Li, L.C. Jia, J. Lei, D.X. Yan, Z.M. Li, Recent progress on segregated polymer composites for functional applications, *Prog Mater Sci.* 155 (2026) 101524, <https://doi.org/10.1016/j.pmatsci.2025.101524>.
- [19] S. Zheng, Y. Wang, X. Wang, H. Lu, Research progress on high-performance electromagnetic interference shielding materials with well-organized multilayered structures, *Materials Today Physics.* (2024) 101330, <https://doi.org/10.1016/j.mtphys.2024.101330>.
- [20] Q. Ren, Y. Wei, X. Li, C. Yu, L. Wang, B. Shen, W. Zheng, Influence of expansion ratio and multilayered gradient structure on the electromagnetic interference shielding performance of lightweight poly (lactic acid)/carbon nanostructures composite foams, *Compos Sci Technol.* 261 (2025) 110992, <https://doi.org/10.1016/j.compscitech.2024.110992>.
- [21] W. Tang, S. Liu, X. Wang, B. Wang, F. Zou, G. Li, X. Liao, Improved electromagnetic interference shielding performance of silicone rubber/CNTs composites by porous structure design via green supercritical CO₂ foaming, *Compos Commun.* 46 (2024) 101808, <https://doi.org/10.1016/j.coco.2023.101808>.
- [22] Y. Yu, J. Fan, G. Xu, J. Zhang, Blends of polydimethylsiloxane-based polyurethane and poly (propylene glycol)-based polyurethane with co-continuous structures: Morphology evolution, synergistic effects and application in strain sensors, *Polymer*. 312 (2024) 127615, <https://doi.org/10.1016/j.polymer.2024.127615>.
- [23] Y. Liu, H. He, G. Tian, Y. Wang, J. Gao, C. Wang, L. Xu, H. Zhang, Morphology evolution to form double percolation polylactide/polycaprolactone/MWCNTs nanocomposites with ultralow percolation threshold and excellent EMI shielding, *Compos Sci Technol.* 214 (2021) 108956, <https://doi.org/10.1016/j.compscitech.2021.108956>.
- [24] Z. Xu, X. Yu, Y. Lu, S. Sun, L. Yang, M. Du, M. Zuo, Q. Zheng, Interfacial distribution of PMMA-grafted MWCNTs in immiscible blends for improved electrical conductivity, *Compos Commun.* 50 (2024) 101997, <https://doi.org/10.1016/j.coco.2024.101997>.
- [25] N. Li, B. Wen, X. Li, S. Yang, G. Yang, S. Ding, Phase structure-induced amplification of interfacial polarization loss for excellent electromagnetic wave absorption, *Chem Eng J.* 488 (2024) 150420, <https://doi.org/10.1016/j.cej.2024.150420>.
- [26] B. Cai, L. Zhou, P.-Y. Zhao, H.-L. Peng, Z.-L. Hou, P. Hu, L.-M. Liu, G.-S. Wang, Interface-induced dual-pinning mechanism enhances low-frequency electromagnetic wave loss, *Nat Commun.* 15 (2024) 3299, <https://doi.org/10.1038/s41467-024-47537-5>.
- [27] Q. Zhang, D. Lan, S. Deng, J. Gu, Y. Wang, J. Ren, G. Wu, Z. Jia, Constructing multiple heterogeneous interfaces in one-dimensional carbon fiber materials for superior electromagnetic wave absorption, *Carbon*. 226 (2024) 119233, <https://doi.org/10.1016/j.carbon.2024.119233>.
- [28] G. Chen, Z.J. Li, L.M. Zhang, Q. Chang, X.J. Chen, X.M. Fan, Q. Chen, H.J. Wu, Mechanisms, design, and fabrication strategies for emerging electromagnetic wave-absorbing materials, *Cell Rep Phys Sci.* 5 (2024) 102097, <https://doi.org/10.1016/j.xcrp.2024.102097>.
- [29] R.P. Modanwal, S.I. Kundalwal, Design and development of low-weight and high-strength electromagnetic shielding nanocomposite in a microwave frequency range, *Polym Compos.* 46 (2025) 3556–3575, <https://doi.org/10.1002/pc.29191>.
- [30] R.P. Modanwal, M. Gupta, S.I. Kundalwal, Effect of tunneling conductivity of CNTs on the EMI shielding effectiveness of nanocomposite in the C-band, *Journal of the Mechanical Behavior of Materials.* 34 (2025), <https://doi.org/10.1515/jmbm-2024-0034>.
- [31] J. Chen, J. Zhang, J. Deng, S. Qiu, J. Zheng, Y. Chen, W. Zhang, S. Huang, A. Chen, C. Lei, Control of graphene nanoplatelets at the interface of the co-continuous polypropylene/polyamides 6 blend under the elongational flow, *Eur Polym J.* 182 (2023) 111703, <https://doi.org/10.1016/j.eurpolymj.2022.111703>.
- [32] E. Pesaranhajiabab, A.K. Pal, A. Rodriguez-Urbe, A.K. Mohanty, M. Misra, Biodegradable polymer blends: studies on performance control through droplet to co-continuous morphology, *ACS Appl Polym Mater.* 4 (2022) 5546–5556, <https://doi.org/10.1021/acsapm.2c00603>.
- [33] L. Zhou, Y. Tian, P. Xu, H. Wei, Y. Li, H.-X. Peng, F. Qin, Effect of the selective localization of carbon nanotubes and phase domain in immiscible blends on tunable microwave dielectric properties, *Compos Sci Technol.* 213 (2021) 108919, <https://doi.org/10.1016/j.compscitech.2021.108919>.
- [34] J.-R. Tao, D. Yang, Y. Yang, Q.-M. He, B. Fei, M. Wang, Migration mechanism of carbon nanotubes and matching viscosity-dependent morphology in co-continuous Poly (lactic acid)/Poly (ε-caprolactone) blend: Towards electromagnetic shielding enhancement, *Polymer.* 252 (2022) 124963, <https://doi.org/10.1016/j.polymer.2022.124963>.
- [35] H. Zhang, H. Jiang, Z.X. Huang, J.P. Qu, Toward high dielectric constant and low dielectric loss nanocomposite via kinetical migration, *Compos Sci Technol.* 221 (2022) 109310, <https://doi.org/10.1016/j.compscitech.2022.109310>.
- [36] X.D. Qi, J.H. Yang, N. Zhang, T. Huang, Z.W. Zhou, I. Kühnert, P. Pötschke, Y. Wang, Selective localization of carbon nanotubes and its effect on the structure and properties of polymer blends, *Prog Polym Sci.* 123 (2021) 101471, <https://doi.org/10.1016/j.progpolymsci.2021.101471>.
- [37] C. Lencar, S. Ramakrishnan, E. Erfanian, U. Sundararaj, The role of phase migration of carbon nanotubes in melt-mixed PVDF/PE polymer blends for high conductivity and EMI shielding applications, *Molecules.* 27 (2022) 933, <https://doi.org/10.3390/molecules27030933>.
- [38] S.S. Hwang, Tensile, electrical conductivity and EMI shielding properties of solid and foamed PBT/carbon fiber composites, *Compos B Eng.* 98 (2016) 1–8, <https://doi.org/10.1016/j.compositesb.2016.05.028>.
- [39] L.Q. Zhang, B. Yang, J. Teng, J. Lei, D.X. Yan, G.J. Zhong, Z.M. Li, Tunable electromagnetic interference shielding effectiveness via multilayer assembly of regenerated cellulose as a supporting substrate and carbon nanotubes/polymer as a functional layer, *J. Mater. Chem. C*. 5 (2017) 3130–3138, <https://doi.org/10.1039/c6tc05516h>.
- [40] M. Sumita, K. Sakata, S. Asai, K. Miyasaka, H. Nakagawa, Dispersion of fillers and the electrical conductivity of polymer blends filled with carbon black, *Polym. Bull.* 25 (1991) 265–271.
- [41] D.K. Owens, R. Wendt, Estimation of the surface free energy of polymers, *J Appl Polym Sci.* 13 (1969) 1741–1747.
- [42] M. Shang, Z. Gao, H. Cheng, C. Wu, B. Shentu, Relationship between microstructure evolution and properties enhancement of carbon nanotubes-filled polybutylene terephthalate/polypropylene blends induced by thermal annealing, *J Appl Polym Sci.* 139 (2022) 51689, <https://doi.org/10.1002/app.51689>.

- [43] J. Palierne, Linear rheology of viscoelastic emulsions with interfacial tension, *Rheol Acta*. 29 (1990) 204–214.
- [44] D. Graebbling, R. Muller, J. Palierne, Linear viscoelastic behavior of some incompatible polymer blends in the melt, Interpretation of data with a model of emulsion of viscoelastic liquids, *Macromolecules*. 26 (1993) 320–329.
- [45] M. Shang, H. Cheng, B. Shentu, Z. Weng, Preparation of PBT/POE-g-GMA/PP ternary blends with good rigidity–toughness balance by core–shell particles, *J Appl Polym Sci*. 137 (2020) 48872.
- [46] P.D. Patel, T. Theofanous, Hydrodynamic fragmentation of drops, *J Fluid Mech*. 103 (1981) 207–223.
- [47] D.K. Lee, S.I. Park, J.K. Lee, N.M. Hwang, A theoretical model for digestive ripening, *Acta Mater*. 55 (2007) 5281–5288.
- [48] L. Utracki, Z. Shi, Development of polymer blend morphology during compounding in a twin-screw extruder. Part I: Droplet dispersion and coalescence—a review, *Polymer Engineering & Science*. 32 (1992) 1824–1833.
- [49] Y. Wang, G. Li, Z. Wan, H. Zhu, Y. Ma, L. Xie, Establishment of dynamic prediction model on morphology development of polypropylene/polyamide 6 blends under in-situ fibrillation flow, *Polymer*. 294 (2024) 126722, <https://doi.org/10.1016/j.polymer.2024.126722>.
- [50] H. Qiao, B. Zheng, G. Zhong, Z. Li, R. Cardinaels, P. Moldenaers, K. Lamnawar, A. Maazouz, C. Liu, H. Zhang, Understanding the rheology of polymer–polymer interfaces covered with Janus nanoparticles: Polymer blends versus particle sandwiched multilayers, *Macromolecules*. 56 (2023) 647–663, <https://doi.org/10.1021/acs.macromol.2c01973>.
- [51] Y. Xia, W. Gao, C. Gao, A review on graphene-based electromagnetic functional materials: electromagnetic wave shielding and absorption, *Adv Funct Mater*. 32 (2022) 2204591, <https://doi.org/10.1002/adfm.202204591>.
- [52] H. Pang, Y. Duan, L. Huang, L. Song, J. Liu, T. Zhang, X. Yang, J. Liu, X. Ma, J. Di, Research advances in composition, structure and mechanisms of microwave absorbing materials, *Compos B Eng*. 224 (2021) 109173, <https://doi.org/10.1016/j.compositesb.2021.109173>.
- [53] M. Qin, L. Zhang, H. Wu, Dielectric loss mechanism in electromagnetic wave absorbing materials, *Adv Sci*. 9 (2022) 2105553, <https://doi.org/10.1002/advs.202105553>.
- [54] S. Zheng, Y. Wang, Y. Zhu, C. Zheng, Recent advances in the construction and properties of carbon-based nanofillers/polymer composites with segregated network structures, *Mater Today Commun*. 36 (2023) 106773, <https://doi.org/10.1016/j.mtcomm.2023.106773>.
- [55] X. Li, D. Niu, P. Xu, W. Yang, P. Ma, High-toughness and antistatic PET/EMAG/CNTs nanocomposites from recycled sources by reactive compatibilization, *Compos Commun*. 47 (2024) 101855, <https://doi.org/10.1016/j.coco.2024.101855>.
- [56] B. Sun, S. Sun, P. He, H.-Y. Mi, B. Dong, C. Liu, C. Shen, Asymmetric layered structural design with segregated conductive network for absorption-dominated high-performance electromagnetic interference shielding, *Chem Eng J*. 416 (2021) 129083, <https://doi.org/10.1016/j.cej.2021.129083>.
- [57] S. Kuester, C. Merlini, G.M.O. Barra, J.C. Ferreira, A. Lucas, A.C. de Souza, B. G. Soares, Processing and characterization of conductive composites based on poly(styrene-*b*-ethylene-*ran*-butylene-*b*-styrene) (SEBS) and carbon additives: A comparative study of expanded graphite and carbon black, *Compos B Eng*. 84 (2016) 236–247, <https://doi.org/10.1016/j.compositesb.2015.09.001>.
- [58] Z.Y. Gao, Q.F. Dong, M.Y. Shang, B.Q. Shentu, C.N. Wu, Microstructure and properties of poly(butylene terephthalate)/poly(ethylene terephthalate) composites based on carbon nanotubes/graphene nanoplatelets hybrid filler systems, *J Appl Polym Sci*. 139 (2022) e51733, <https://doi.org/10.1002/app.51733>.
- [59] J. Cai, L. Wang, H.J. Duan, Y. Zhang, X.Y. Wang, G. Wan, Z.L. Zhong, Porous polyamide 6/carbon black composite as an effective electromagnetic interference shield, *Polym Int*. 71 (2022) 247–254, <https://doi.org/10.1002/pi.6311>.
- [60] D.W. Xu, W.H. Chen, P.J. Liu, Enhanced electromagnetic interference shielding and mechanical properties of segregated polymer/carbon nanotube composite via selective microwave sintering, *Compos Sci Technol*. 199 (2020) 108355, <https://doi.org/10.1016/j.compscitech.2020.108355>.
- [61] X.Y. Ji, D.Y. Chen, J.B. Shen, S.Y. Guo, Flexible and flame-retarding thermoplastic polyurethane-based electromagnetic interference shielding composites, *Chem, Eng J*. 370 (2019) 1341–1349, <https://doi.org/10.1016/j.cej.2019.03.293>.
- [62] J. Chen, X. Liao, S.J. Li, W. Wang, F.M. Guo, G.X. Li, A promising strategy for efficient electromagnetic interference shielding by designing a porous double-percolated structure in MWCNT/polymer-based composites, *Compos A Appl Sci Manuf*. 138 (2020) 106059, <https://doi.org/10.1016/j.compositesa.2020.106059>.
- [63] R. Ravindren, S. Mondal, K. Nath, N.C. Das, Synergistic effect of double percolated co-supportive MWCNT-CB conductive network for high-performance EMI shielding application, *Polym Adv Technol*. 30 (2019) 1506–1517, <https://doi.org/10.1002/pat.4582>.
- [64] K. Zhan, T. Elder, Y. Peng, Enhancing Polypropylene/Polyethylene Blend Performance Through Compatibilization for A Sustainable Future: A Mini Review Focusing on Establishing Bio-Derived Filler Based Hybrid Compatibilizer System, *Macromol Rapid Commun*. 46 (2025), <https://doi.org/10.1002/marc.202400724>.
- [65] J. Chen, C. Rong, T. Lin, Y. Chen, J. Wu, J. You, H. Wang, Y. Li, Stable Co-Continuous PLA/PBAT Blends Compatibilized by Interfacial Stereocomplex Crystallites: Toward Full Biodegradable Polymer Blends with Simultaneously Enhanced Mechanical Properties and Crystallization Rates, *Macromolecules*. 54 (2021) 2852–2861, <https://doi.org/10.1021/acs.macromol.0c02861>.
- [66] R.W. Clarke, T. Sandmeier, K.A. Franklin, D. Reich, X. Zhang, N. Vengallur, T. K. Patra, R.J. Tannenbaum, S. Adhikari, S.K. Kumar, T. Rovis, E.Y.X. Chen, Dynamic crosslinking compatibilizes immiscible mixed plastics, *Nature*. 616 (2023) 731, <https://doi.org/10.1038/s41586-023-05858-3>.
- [67] M. Shang, H. Cheng, B. Shentu, Z. Weng, Preparation of PBT/POE-g-GMA/PP ternary blends with good rigidity–toughness balance by core–shell particles, *J Appl Polym Sci*. 137 (2020) 48872, <https://doi.org/10.1002/app.48872>.